

Q-1

a) Convolutional Neural Network diagram -

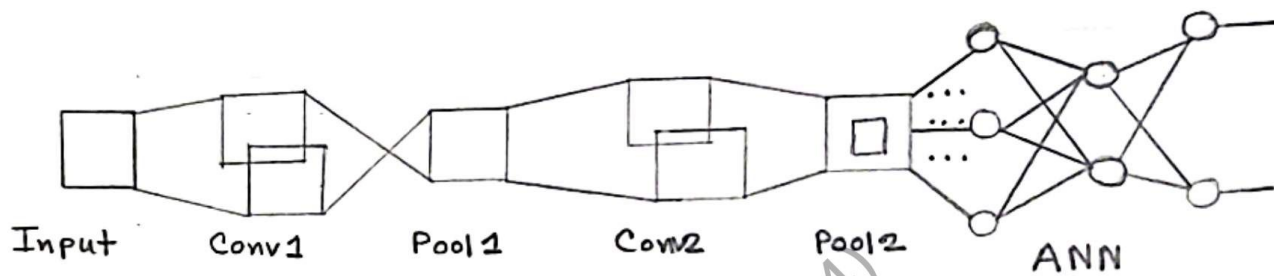


Fig - Convolutional Operation.

b)

$$\text{Input, } 4 \times 4 = \begin{bmatrix} 1 & 2 & 3 & 4 \\ -2 & 3 & 1 & 1 \\ 3 & -3 & -2 & 1 \\ 2 & 1 & 1 & 4 \end{bmatrix}$$

$$\text{Convolutional matrix} = \begin{bmatrix} 2 & 1 & 5 \\ 1 & 3 & 1 \\ 4 & 2 & 1 \end{bmatrix} ; \text{kernel} = 3 \times 3$$

$$\text{Stride} = 1$$

$$\text{Average pooling} = 2 \times 2$$

So,

$$\begin{bmatrix} 1 & 2 & 3 \\ -2 & 3 & 1 \\ 3 & -3 & -2 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 5 \\ 1 & 3 & 1 \\ 4 & 2 & 1 \end{bmatrix}$$

$$= 2 + 2 + 15 - 2 + 9 + 1 + 12 - 6 - 2$$

$$= 31$$

$$\begin{bmatrix} 2 & 3 & 4 \\ 3 & 1 & 1 \\ -3 & -2 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 5 \\ 1 & 3 & 1 \\ 4 & 2 & 1 \end{bmatrix} = 4 + 3 + 20 + 3 + 3 + 1 - 12 - 4 + 1$$

$$= 12$$

$$\begin{bmatrix} 3 & 1 & 1 \\ -3 & -2 & -1 \\ 1 & 1 & 4 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 5 \\ 1 & 3 & 1 \\ 4 & 2 & 1 \end{bmatrix} = 3 + 1 + 5 - 3 - 6 + 1 + 4 + 2 + 4$$

$$= 11$$

$$\begin{bmatrix} -2 & 3 & 1 \\ 3 & -3 & -2 \\ 2 & 1 & 1 \end{bmatrix} * \begin{bmatrix} 2 & 1 & 5 \\ 1 & 3 & 1 \\ 4 & 2 & 1 \end{bmatrix} = -4 + 3 + 5 + 3 - 9 - 2 + 8 + 2 + 1$$

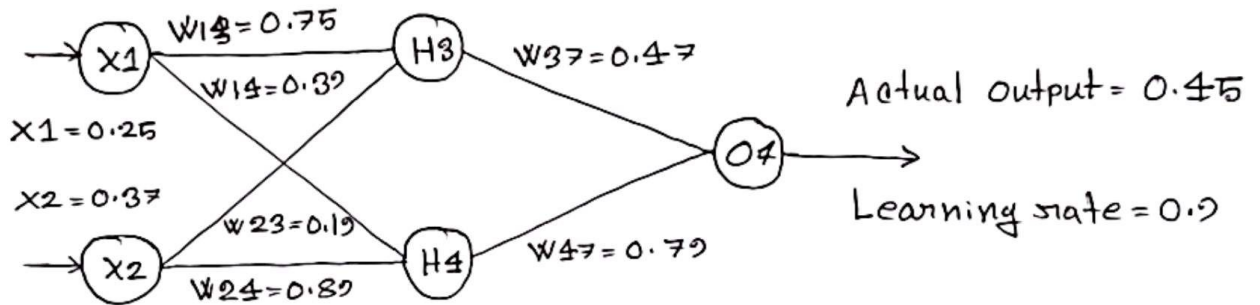
$$= 7$$

$$\therefore \text{Conv1} = \begin{bmatrix} 3 & 1 & 12 \\ 7 & & 11 \end{bmatrix}$$

$$\therefore \text{Average Pool1} = 17$$

Ans

Q-2



Epoch 1

Forward pass -

For H3,

$$\begin{aligned} \text{Input, H3} &= X_1 \times W_{13} + X_2 \times W_{23} \\ &= 0.25 \times 0.75 + 0.37 \times 0.19 \\ &= 0.2578 \end{aligned}$$

$$\text{Output, H3} = \frac{1}{1 + e^{-0.2578}} = 0.56$$

For H4,

$$\begin{aligned} \text{Input, H4} &= X_1 \times W_{14} + X_2 \times W_{24} \\ &= 0.25 \times 0.39 + 0.37 \times 0.89 \\ &= 0.4268 \end{aligned}$$

$$\text{Output, H4} = \frac{1}{1 + e^{-0.4268}} = 0.61$$

For O7,

$$\begin{aligned} \text{Input, O7} &= H_3 \times W_{37} + H_4 \times W_{47} \\ &= 0.56 \times 0.47 + 0.61 \times 0.79 \\ &= 0.7451 \end{aligned}$$

$$\text{Output, } O_7 = \frac{1}{1 + e^{-0.7451}} = 0.68$$

$$\begin{aligned}\therefore \text{Error} &= 0.68 - 0.45 \\ &= 0.23\end{aligned}$$

Backward Pass -

For O_7 ,

$$\begin{aligned}S_7 &= Y_7(1 - Y_7)(t_7 - Y_7) \\ &= 0.68(1 - 0.68)(0.45 - 0.68) \\ &= -0.05\end{aligned}$$

$$\begin{aligned}\Delta W_{37} &= \eta S_7 Y_3 \\ &= 0.9 \times (-0.05)(0.56) \\ &= -0.0252\end{aligned}$$

$$\begin{aligned}\therefore W_{37}(\text{new}) &= \Delta W_{37} + W_{37}(\text{old}) \\ &= -0.0252 + 0.47 \\ &= 0.44\end{aligned}$$

$$\begin{aligned}\Delta W_{47} &= \eta S_7 Y_4 \\ &= 0.9 \times (-0.05)(0.61) \\ &= -0.02745\end{aligned}$$

$$\begin{aligned}\therefore W_{47}(\text{new}) &= \Delta W_{47} + W_{47}(\text{old}) \\ &= -0.02745 + 0.79 \\ &= 0.76\end{aligned}$$

For H_3 ,

$$\begin{aligned} S_3 &= y_3(1-y_3)(w_{37} \times S_7) \\ &= 0.56(1-0.56)(0.47 \times \overset{-0.05}{0.68}) \\ &= -0.006 \end{aligned}$$

$$\begin{aligned} \Delta w_{13} &= \eta S_3 y_1 \\ &= 0.2(-0.006)(0.25) \\ &= -0.00135 \end{aligned}$$

$$\begin{aligned} \therefore w_{13}(\text{new}) &= \Delta w_{13} + w_{13}(\text{old}) \\ &= -0.00135 + 0.75 \\ &= 0.748 \end{aligned}$$

$$\begin{aligned} \Delta w_{23} &= \eta S_3 y_2 \\ &= 0.2(-0.006)(0.37) \\ &= -0.001728 \end{aligned}$$

$$\begin{aligned} \therefore w_{23}(\text{new}) &= \Delta w_{23} + w_{23}(\text{old}) \\ &= -0.001728 + 0.19 \\ &= 0.188 \end{aligned}$$

For H_4 ,

$$\begin{aligned} S_4 &= y_4(1-y_4)(w_{47} \times S_7) \\ &= 0.61(1-0.61)(0.72 \times -0.05) \\ &= -0.009 \end{aligned}$$

$$\begin{aligned} \Delta w_{14} &= \eta S_4 y_1 \\ &= 0.2 \times (-0.009) \times (0.25) \\ &= -0.002 \end{aligned}$$

$$\therefore W_{14}(\text{new}) = \Delta W_{14} + W_{14}(\text{old})$$

$$= -0.002 + 0.39$$

$$= 0.388$$

$$\Delta W_{24} = \eta \delta_4 y_2$$

$$= 0.9 (-0.009)(0.37)$$

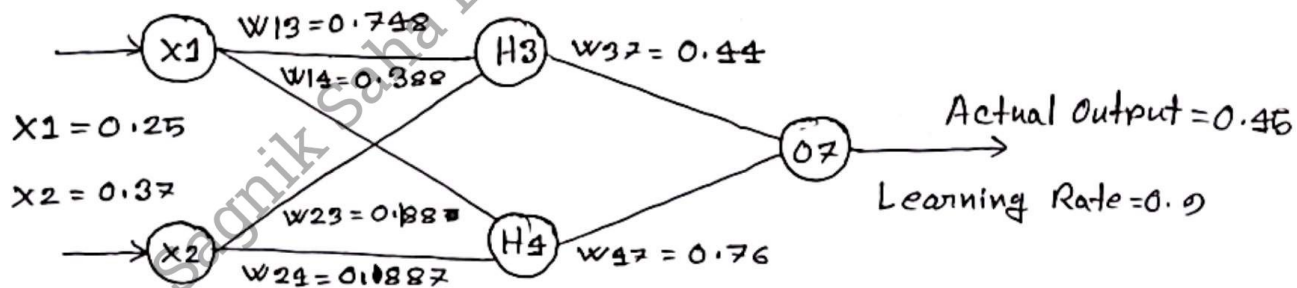
$$= -0.003$$

$$\therefore W_{24}(\text{new}) = \Delta W_{24} + W_{24}(\text{old})$$

$$= -0.003 + 0.89$$

$$= 0.887$$

$\therefore$ Updated Weight -



Epoch 2

Forward Pass -

For H_3 ,

$$\text{Input, } H_3 = X_1 \times W_{13} + X_2 \times W_{23}$$

$$= 0.25 \times 0.748 + 0.37 \times 0.188$$

$$= 0.25656$$

$$\text{Output, } H3 = \frac{1}{1 + e^{-0.25656}} = 0.56$$

For $H4$,

$$\begin{aligned} \text{Input, } H4 &= X1 \times W14 + X2 \times W24 \\ &= 0.25 \times 0.388 + 0.37 \times 0.887 \\ &= 0.42519 \end{aligned}$$

$$\text{Output, } H4 = \frac{1}{1 + e^{-0.42519}} = 0.6$$

For $O7$,

$$\begin{aligned} \text{Input, } O7 &= H3 \times W37 + H4 \times W47 \\ &= 0.56 \times 0.44 + 0.6 \times 0.76 \\ &= 0.7024 \end{aligned}$$

$$\text{Output, } O7 = \frac{1}{1 + e^{-0.7024}} = 0.67$$

$$\therefore \text{Error} = 0.67 - 0.45$$

$$= 0.22$$

Ans